Assignment # 2

Artificial Intelligence

Total Marks: 75 **Submission Date:** 12-04-2026

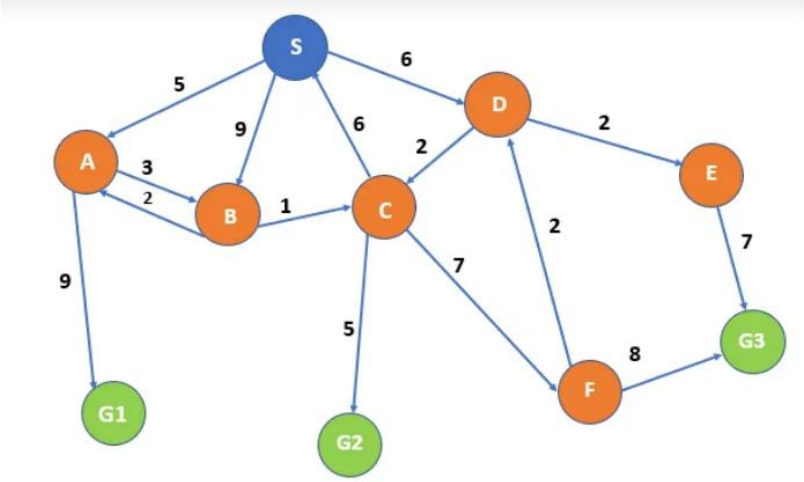
Name_____ **Roll No.** _____

Instructions:

I. Assignment should be hand written.
II. You should create your own solution according to your logic. Any similarity with other student and internet will be consider as cheating and your assignment will be marked as ZERO.

Question #1: Find shortest distance from Lugoj to Bucharest using **UCS** and **A*** algorithm in following map. Show all steps of state space and queue processing. **[20 marks]**

There are multiple goals(G1, G2,and G3) in the following graph starting from S node.



Question # 2: **[10 marks]**

Create a search tree for the 8-puzzle problem, solve using **Best First search** algorithm. In this tree, $g(n)$ represents the number of steps from the initial state, while $h(n)$ represents the number of misplaced tiles.

Start state

1	2	3
4	8	5
7	6	

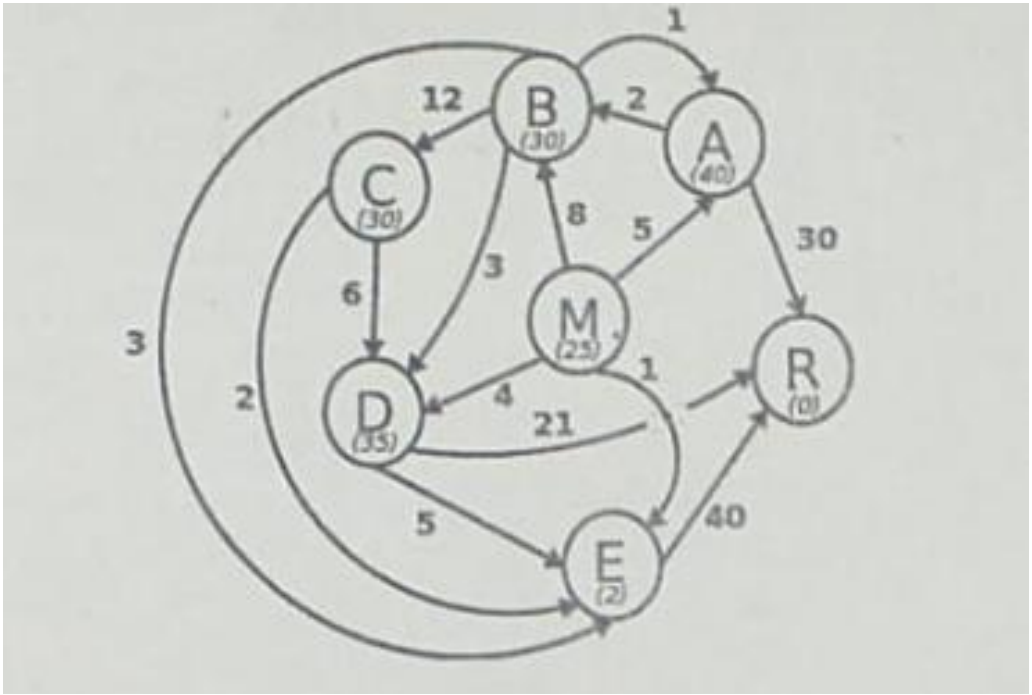
$\longrightarrow$

Goal state

1	2	3
4	5	6
7	8	

Note: keep in view that one node can be explored only once

Question # 3: consider the search problem below with start state **M** and goal state **R.** the number on the edges show path cost while the number inside each node show heuristic value **[30+15=45 Marks]**



I. a) Apply A* algorithm on the above graph by showing the frontier, order of expansion and visited nodes. Also show the path from start to goal state. Break the ties based on the alphabetical order.
b) Check if above graph is admissible and consistent. Show all steps.

II. Apply IDA* on the above graph and show f-bound and pruned list for each iteration